



Year 11 Mathematics 2 Unit
Preliminary HSC Assessment Task 2
Term 3, Week 3, 2020

Name: _____

Teacher: _____

Wednesday 5th August 2020

Setter: CHG

- Answer all questions on the lined paper provided.
- Start each question on a new page.
- Board approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- For any multiple choice question, **write the letter** that corresponds to your answer on your writing paper.
- Total possible mark: **32**
- Time allowed: **45 minutes + 2 minutes reading time**

Question 1 (8 marks)

Marks

(a) If $\tan \theta = \sqrt{3}$ and $\sin \theta < 0$, which value for $0 \leq \theta \leq 360^\circ$ satisfies these conditions. **1**

(A) 60°

(B) 120°

(C) 240°

(D) 400°

(b) A formula for the temperature of water after t hours is $T(t) = 100 \times \left(\frac{1}{2}\right)^t$. **1**

Using this formula calculate the time taken for the water to cool to 10°C .
Leave your answer in hours correct to 2 decimal places.

(c) Solve these equations for $[0^\circ, 360^\circ]$:

(i) $\sin 2x = \frac{1}{\sqrt{2}}$ **2**

(ii) $8\sin x = \operatorname{cosec}^2 x$ **2**

(iii) $2\sin^2 x + \cos x = 1$ **2**

End of Question 1

Question 2 starts on a new page

Question 2 (8 marks) *Start a new page***Marks**

(a) Differentiate: $y = 6x^2 - 4$ **1**

(A) $6x^2$

(B) $12x$

(C) $12x - 1$

(D) $2x^3 - 4x$

Differentiate with respect to x , leaving your answer in simplest positive integer or surd form.

(b) $y = -\frac{2}{x^2}$ **1**

(c) $y = \sqrt{9x^2 - 16}$ **2**

(d) $y = (2x^2 + 5)(3x - 5)^4$ (Leaving your answer in fully factorised form) **2**

(e) $y = \frac{4x^2 - 3}{x + 3}$ **2**

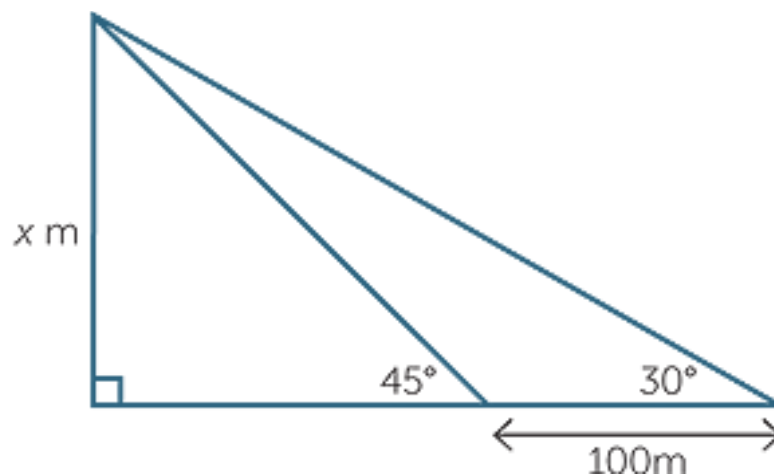
End of Question 2

Question 3 starts on a new page

Question 3 (8 marks) *Start a new page*

Marks

- (a) If $\cos \theta = -0.5$, which values for $0 \leq \theta \leq 360^\circ$ satisfy these conditions? **1**
- (A) $120^\circ, 240^\circ$
- (B) $150^\circ, 330^\circ$
- (C) $210^\circ, 330^\circ$
- (D) $240^\circ, 300^\circ$
- (b) Given that $\log_b 3 = 1.2$, find the value of $\log_b 3b^2$. **1**
- (c) Find the equation of the normal to the curve $y = 2\sqrt{x}$ at the point (4,4). **2**
- (d) Prove: $\frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$ **2**
- (e) Calculate the value of x in the diagram below, answer correct to 2 decimal places. **2**



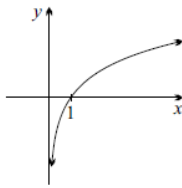
End of Question 3

Question 4 starts on a new page

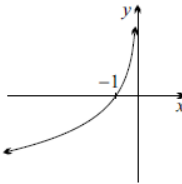
- (a) Which graph represents $y = -\log(x)$?

1

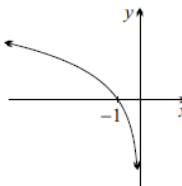
(A)



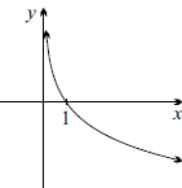
(B)



(C)



(D)



- (b) Tim ran 2km on a bearing of 040° , turned and ran 3km on a bearing of 150° . Calculate the distance he is now from his starting position to the nearest km.

1

- (c) Find the x -coordinates on the curve $y = 2x^3 - 3x^2 - 12x + 10$ where the tangent is parallel to the x -axis.

2

- (d) Differentiate $f(x) = x^2 - 4x$ using first principles.

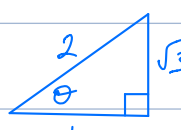
2

- (e) Prove $\log_2 5$ is irrational.

2

End of Assessment

2020 11 Advanced AT2 Solutions

i) a)  $\sin \theta = \frac{\sqrt{3}}{2}$ $\angle_{\text{related}} = 60^\circ$ $\theta = 180 + 60 = 240^\circ$ (C) ✓

b) $10 = 100 \left(\frac{1}{2}\right)^t$
 $0.1 = \left(\frac{1}{2}\right)^t$
 $t = \frac{\log(0.1)}{\log(0.5)}$
 $\approx 3.32 \text{ hours}$ ✓

c) i) $\sin 2x = \frac{1}{\sqrt{2}} \quad 0 \leq x \leq 360$
 $2x = 45, 135, 405, 495 \quad 0 \leq 2x \leq 720$
 $x = 22.5^\circ, 67.5^\circ, 202.5^\circ, 247.5^\circ$ ✓✓
 (1) 45 and 135°

ii) $8 \sin x = \operatorname{cosec}^2 x$
 $8 \sin x = \frac{1}{\sin^2 x}$
 $8 \sin^3 x = 1$
 $\sin^3 x = \frac{1}{8}$ (1)
 $\sin x = \sqrt[3]{\frac{1}{8}}$
 $\sin x = \frac{1}{2}$
 $x = 30, 150^\circ$ ✓✓

iii) $2 \sin^2 x + \cos x = 1$
 $2(1 - \cos^2 x) + \cos x - 1 = 0$
 $2 - 2\cos^2 x + \cos x - 1 = 0$
 $-2\cos^2 x + \cos x + 1 = 0$
 $2\cos^2 x - \cos x - 1 = 0$ (1)
 $(\cos x - 1)(2\cos x + 1) = 0$

$\therefore \cos x - 1 = 0$ or $2\cos x + 1 = 0$
 $\cos x = 1$ $\cos x = -\frac{1}{2}$
 $x = 0, 360^\circ$ $x = 120^\circ, 240^\circ$
 $\therefore x = 0^\circ, 120^\circ, 240^\circ, 360^\circ$ ✓✓

(2) a) (B) $y = 6x^2 - 4$
 $y' = 12x$ ✓

b) $y = 2x^{-2}$
 $y' = \frac{4}{x^3}$ ✓

c) $y = (9x^2 - 16)^{1/2}$
 $y' = \frac{1}{2}(18x)(9x^2 - 16)^{-1/2}$ (1)
 $= \frac{18x}{2\sqrt{9x^2 - 16}}$
 $= \frac{9x}{\sqrt{9x^2 - 16}}$ ✓✓

$$\begin{aligned} \textcircled{2} d) \quad y &= (2x^2+5)(3x-5)^4 \\ \textcircled{1} \quad &= 4x(3x-5)^4 + 12(2x^2+5)(3x-5)^3 \\ &= (3x-5)^3 (4x(3x-5) + 24x^2+60) \\ &= (3x-5)^3 (12x^2 - 20x + 24x^2 + 60) \\ &= 4(3x-5)^3 (9x^2 - 5x + 15) \quad \checkmark \checkmark \end{aligned}$$

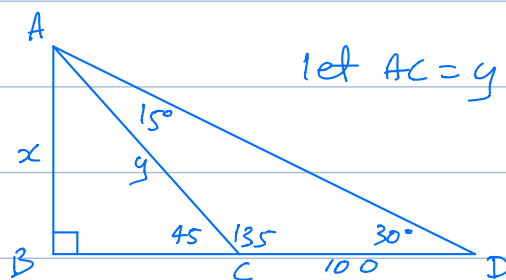
$$\begin{aligned} e) \quad y &= \frac{4x^2-3}{x+3} \\ &= \frac{8x(x+3) - (4x^2-3)}{(x+3)^2} \quad \textcircled{1} \\ &= \frac{8x^2 + 24x - 4x^2 + 3}{(x+3)^2} \\ &= \frac{4x^2 + 24x + 3}{(x+3)^2} \quad \checkmark \checkmark \end{aligned}$$

$$\begin{aligned} 3a) \quad \textcircled{A} \angle_{\text{related}} &= 60^\circ \\ \theta &= 180-60, 180+60 \\ &= 120^\circ, 240^\circ \quad \checkmark \end{aligned}$$

$$\begin{aligned} b) \quad &= \log_b 3b^2 \\ &= \log_b 3 + 2\log_b b \\ &= 1.2 + 2 \\ &= 3.2 \quad \checkmark \end{aligned}$$

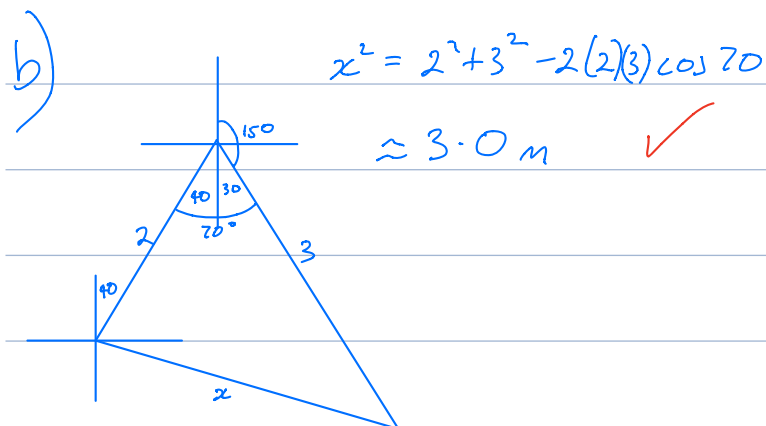
$$\begin{aligned} \textcircled{3} c) \quad y &= 2x^{\frac{1}{2}} \\ y' &= 2\left(\frac{1}{2}\right)x^{-\frac{1}{2}} \\ &= \frac{1}{\sqrt{x}} \\ \text{using} \quad M_T &= \frac{1}{\sqrt{4}} = \frac{1}{2} \quad \textcircled{1} \text{ and } (4, 4) \\ y-4 &= \frac{1}{2}(x-4) \\ y &= \frac{x}{2} + 2 \quad \checkmark \checkmark \end{aligned}$$

$$\begin{aligned} d) \quad &\text{Prove} \\ &\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \sec \theta \\ \text{LHS} &= \frac{1}{\sec \theta - \tan \theta} - \tan \theta \\ &= \frac{\sec \theta + \tan \theta}{\sec^2 \theta - \tan^2 \theta} - \tan \theta \quad \textcircled{1} \\ &= \sec \theta + \tan \theta - \tan \theta \\ &= \sec \theta \\ &= \text{RHS} \quad \checkmark \end{aligned}$$

e) 

$$\begin{aligned} \frac{y}{\sin 30} &= \frac{100}{\sin 15} \quad \approx 193 \quad \textcircled{1} \\ y &= \frac{100 \sin 30}{\sin 15} \\ \sin 45 &= \frac{x}{\frac{100 \sin 30}{\sin 15}} \\ x &= (\sin 45) \frac{100 \sin 30}{\sin 15} \\ &\approx 136.6 \text{ m} \quad \checkmark \checkmark \end{aligned}$$

(4) a) (D) ✓



c) $y = 2x^3 - 3x^2 - 12x + 10$
 $y' = 6x^2 - 6x - 12$
let $y' = 0$

$$0 = 6x^2 - 6x - 12$$
$$= 6(x^2 - x - 2)$$
$$= 6(x+1)(x-2) \quad \textcircled{1}$$

$$\therefore x+1=0 \quad \text{or} \quad x-2=0$$
$$x=-1 \quad \quad \quad x=2$$

✓✓

d)

$$f(x) = x^2 - 4x$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4(x+h) - x^2 + 4x}{h}$$
$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - x^2 + 4x}{h}$$
$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h}$$
$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 4)}{h}$$
$$= \lim_{h \rightarrow 0} 2x + h - 4$$
$$= 2x - 4$$

① or $-(x-4)$ ✓✓

e) Assume $\log_2 5$ is rational.
 $\log_2 5 = \frac{m}{n}$

where m & n are integers,
 $n \neq 0$ and have no common factors.

$$2^{\frac{m}{n}} = 5$$
$$2^m = 5^n \quad \textcircled{1}$$

As 2 is even so is 2^m

As 5 is odd so is 5^n

$$\therefore 2^m \neq 5^n \quad \checkmark \checkmark$$

This is a contradiction
so the original
assumption is wrong.
 $\therefore \log_2 5$ is irrational

2 marks for saying one
is even and one is
odd $\therefore$ can't be
true